

①
0.777
X4485986

תרגיל 4 סיכום

307

$$R = 2\Omega$$

$$P = 450W$$

$$P = I^2 \cdot R$$

$$450 = I^2 \cdot 2$$

$$I = 15A$$

$$P = V \cdot I$$

$$450 = V \cdot 15$$

$$V = 30V$$

$$V = 230V$$

$$P = 2000W$$

$$P = V \cdot I$$

$$2000 = 230 \cdot I$$

$$I = 8.695A$$

$$R = \frac{V}{I}$$

$$R = 230 / 8.695$$

$$R = 26.451\Omega$$

$$V = 115$$

$$P = \frac{115^2}{26.451} = 500W$$

$$P = \frac{V^2}{R}$$

3

$$V = 20$$

$$E = 21.6kJ$$

.1c

2.712

Expt
t=0.5s

2nd expt
P=1W

$$E = P \cdot t$$

$$t = 5400s$$

$$21600 = P \cdot 5400$$

$$P = 4W$$

$$R = \frac{V}{I} \quad .P$$

$$P = V \cdot I$$

$$4 = 20 \cdot I$$

$$I = \frac{1}{5}A$$

$$R = 100\Omega$$

$$I = \frac{1}{5}A \quad .d$$

$$V = 40 - 20 = 20V \quad .4$$

$$P = \frac{V^2}{R}$$

$$P = 200W$$

$$200 = \frac{400}{R}$$

$$R = \frac{400}{200} = 2\Omega$$

$$I = \frac{V}{R}$$

$$I = \frac{20}{2} = 10A$$

$$E = P \cdot t \quad .d$$

$$I = 4A$$

$$t = 7200$$

$$V = 20$$

$$P = 20 \cdot 4 = 80 \quad I^2 \cdot R = 32$$

$$E = 80 \cdot 7200 = 576kJ$$

2

307

סדרה 4

128N

$$I = \frac{V}{R}$$

$$I = \frac{2 \cdot 128N}{R}$$

$$V = 9$$

$$R_{eq} = 3000 \Omega$$

$$I = \frac{9}{3000} = 0.003$$

$$I = 3mA$$

$$V_i = I \cdot R_i$$

$$V_i = I \cdot R_i$$

$$V_1 = 0.003 \cdot 1000 = 3V$$

$$V_2 = 0.003 \cdot 240 = 0.72V$$

$$V_3 = 0.003 \cdot 260 = 0.78V$$

$$V_4 = 0.003 \cdot 1500 = 4.5V$$

$$P = V \cdot I$$

$$P_1 = 0.009$$

$$P_2 = 0.00216$$

$$P_3 = 0.00234$$

$$P_4 = 0.0135$$

$$P = 0.027 W$$

$$\frac{R_1}{R_1 + R_2 + R_3 + R_4} = \frac{100}{3000} = 3.3\%$$

$$I = \frac{V}{R}$$

$$\frac{1}{R_{eq}} = \frac{1}{2000} + \frac{1}{4000} + \frac{1}{4000} = \frac{2}{2000} = \frac{1}{1000}$$

$$R_{eq} = 1000 \Omega$$

$$I = \frac{24}{1000} = 24mA$$

$$24V = 24 \cdot 1000 = 24000$$

$$24V \rightarrow 24 \cdot 1000 = 24000$$

$$I_1 = \frac{24}{2000} = 12mA$$

$$I_2 = \frac{24}{4000} = 6mA$$

$$I_3 = 6mA$$

$$P_i = V \cdot I_i$$

$$P_1 = 24 \cdot 0.012$$

$$P_2 = P_3 = 24 \cdot 0.006$$

$$P_1 + P_2 + P_3 = 0.576W$$

$$\frac{\frac{1}{2000}}{\frac{1}{2000} + \frac{1}{4000} + \frac{1}{4000}} = \frac{1}{2} = 50\%$$

$$\frac{1}{1000}$$

③

תנאי 4 פתרון

3'07

2480 7.1/14

22) סך נח

$$V_i = I \cdot R_i$$

$$V_1 = V_T \cdot \frac{R_1}{R_1 + R_2}$$

$$V_{1,2} = \frac{1333.3}{9499.3} \cdot 12 = 1.684V$$

$$V_1 = V_2 = 1.684$$

$$V_3 = 12 \cdot \frac{3000}{9499.3} = 3.789V$$

$$V_4 = V_5 = \frac{12 \cdot 2500}{9499.3} = 3.158V$$

$$V_6 = \frac{12 \cdot 2666}{9499.3} = 3.367V$$

$$R_{1,2} = \frac{4000 \cdot 2000}{4000 + 2000} \quad \text{הקטנה המשותפת}$$

$$= 1333.3$$

$$R_{4,5} = \frac{5000^2}{10000} = 2500\Omega$$

$$R_T = 1333.3 + 2500 + 3000 = 2666.3$$

$$= 9499.33$$

$$I = \frac{V}{R}$$

$$I = \frac{12}{9499.33} = 1.263mA$$

: סך נח

: 127007

$$P = V \cdot I$$

$$P_{1,2} = 2666.3$$

$$P = 0.001263 \cdot 12 = 0.0151 =$$

$$1.5.1mA$$

$$I_1 = I_T \cdot \frac{R_2}{R_1 + R_2}$$

: 32) סך נח

$$I_1 = 2 \cdot \frac{1263 \cdot 2000}{6000} = 421A$$

$$I_2 = 842A$$

$$I_T = I_6 = I_3 = 1263A$$

$$I_5 = I_4 = 1263/2 = 631.5A$$

4

4 7 5 0 12 12

2'07

3 4180

$$V_i = I \cdot R_i$$

:32) 67 121

$$V_1 = 2 \cdot 4 = 8 \text{ V}$$

$$V_2 = V_3 = 2 \cdot 2 = 4 \text{ V}$$

~~V4~~

$$V_4 = V_5 = V_6 = 2 \cdot 1 = 2 \text{ V}$$

$$V_7 = 2 \cdot 3 = 6 \text{ V}$$

$$R_{2,3} = \frac{4 \text{ ohms} \cdot 4 \text{ ohms}}{4 \text{ ohms} + 4 \text{ ohms}} = 2 \text{ ohms}$$

$$= 2000 \Omega$$

$$\frac{1}{R_{4,5,6}} = \frac{1}{6 \text{ ohms}} + \frac{1}{3 \text{ ohms}} + \frac{1}{2 \text{ ohms}} = \frac{5}{6 \text{ ohms}}$$

$$R_{4,5,6} = 1.2 \text{ ohms}$$

$$R_T = 2 \text{ ohms} + 1 \text{ ohms} + 4 \text{ ohms} + 3 \text{ ohms} = 10 \text{ ohms}$$

$$I = \frac{V}{R} \quad : 11.5 \text{ 121}$$

$$I_T = \frac{20}{10} = 2 \text{ A}$$

:32) 67 121

$$P = V \cdot I$$

:11.5 121

$$P = 20 \cdot 2 = 40 \text{ W}$$

$$I_1 = I_T \cdot \frac{R_2}{R_1 + R_2}$$

:32) 67 121

$$I_T = I_1 = I_7 = I_{2,3} = I_{4,5,6} = 2$$

$$I_2 = \frac{2 \cdot 4}{8} = 1 \Omega$$

$$I_3 = 1 \Omega$$

$$I_4 = \frac{2 \cdot R_{5,6}}{R_4 + R_{5,6}} = \frac{2 \cdot \frac{6}{5}}{4 + \frac{6}{5}} = 0.667 \Omega$$

$$I_5 = I_4 \cdot 2 = 0.667 \Omega$$

$$I_6 = I_4 \cdot 3 = 1 \Omega$$

Trans.

(5)

תרגיל פיצוק 4

סוף

פ

מנה A נסירה וטין טאין
 ציר מקבילי- A טאין מנחל חמל.
 בל מלף מנה לא תעלג +
 און צרם ואין הספק.

$$V = 100V$$

$$\underline{\underline{100V}}$$

$$P = 250W$$

$$P = \frac{V^2}{R}$$

$$250 = \frac{1000}{R}$$

$$R_A = R_B = R_C = R_D = 40\Omega$$

~~Transistor~~

$$I_T = 2.5A$$

$$R_{BCD} = \frac{40}{3} = 13.333\Omega$$

$$R_A = 40\Omega$$

$$R_T = 13.333 + 40 = 53.333$$

$$I_A = \frac{2.5 \cdot 13.333}{40 + 13.333}$$

$$I_i = \frac{V}{R_i}$$

$$I_A = \frac{100}{53.33} = 1.875A$$

$$I_B + I_C + I_D = 1.875A$$

$$I_B = I_C = I_D = 0.625A$$

$$P_A = 3.515 \cdot 40 = 140.625W$$

$$P_A = 3.515 \cdot 40 = 140.625W$$

$$P_B = P_C = P_D = 0.625^2 \cdot 40 = 15.625W$$